



Bias and Fairness: definitions and measurements

Ethics & Bias in AI: Lecture 3

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Lecture overview

- Types and sources of bias in ML systems
- How to define fairness
- How do we measure fairness?
- Use with caution: considerations / limitations of error metrics

Where does bias come from?



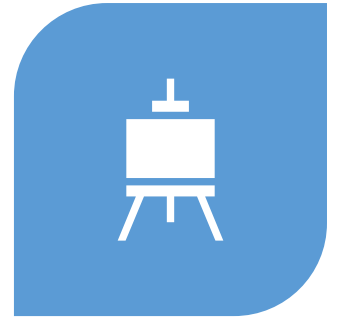
DATA



ANNOTATION



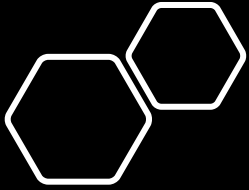
MODEL



DESIGN



Definitions of Bias



Bias recap

Bias is the basis of machine learning:

- The whole point is to discover patterns that differentiates between different groups

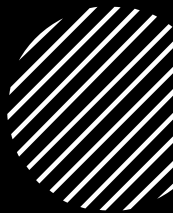
Algorithmic bias:

- *Unjust, unfair, or prejudicial treatment of people related to race, income, sexual orientation, religion, gender, and other characteristics historically associated with discrimination and marginalization, when and where they manifest in algorithmic systems or algorithmically aided decision-making*

(Chang et al. 2019).



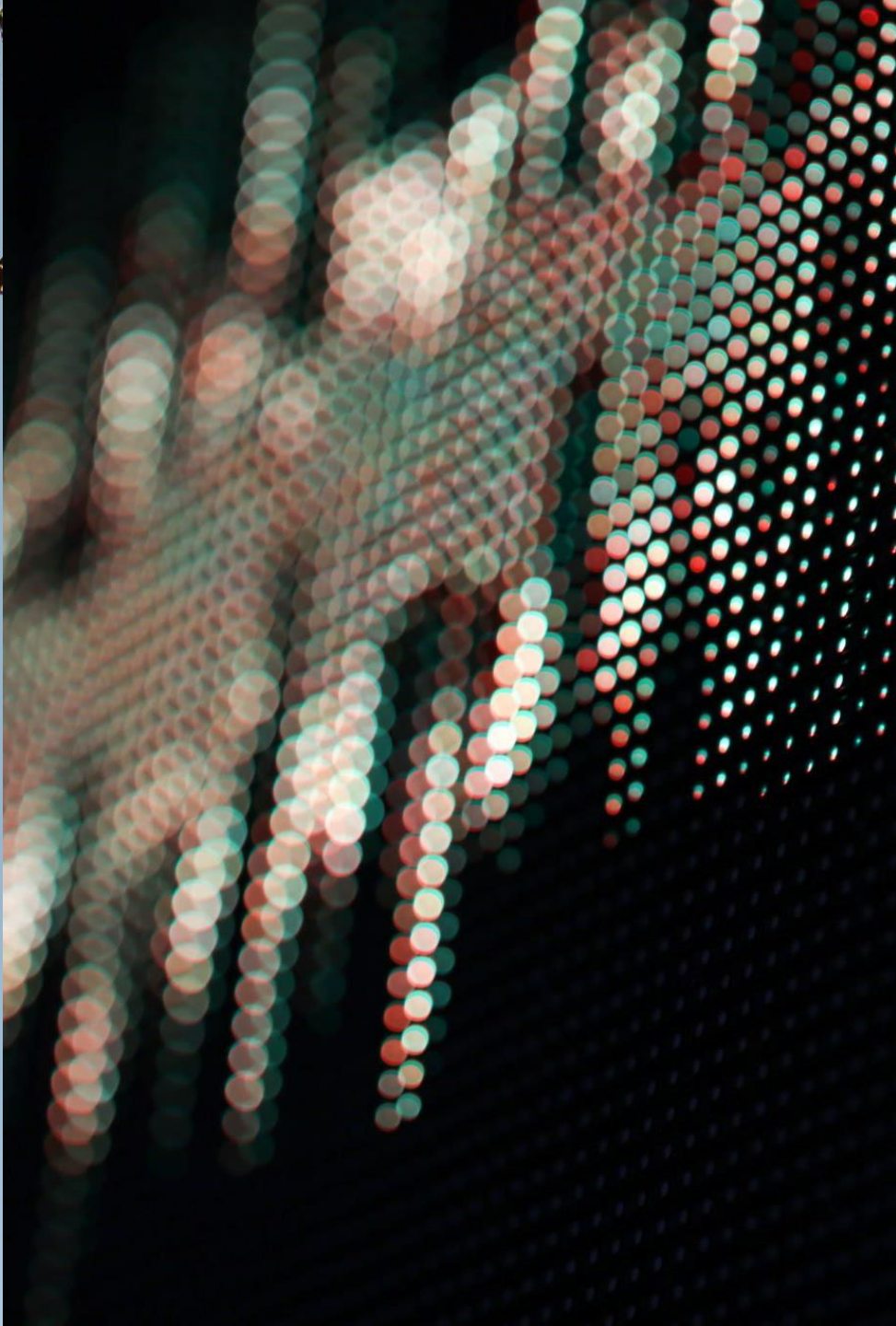
Defining bias in computational & algorithmic systems



Certain aspects of a dataset are more heavily weighted and/or represented than others. This can cause low accuracy and skewed outcomes.

Different performance for diverse groups of people.

Computational systems that systematically and unfairly discriminate against certain individuals or groups of individuals in favor of others

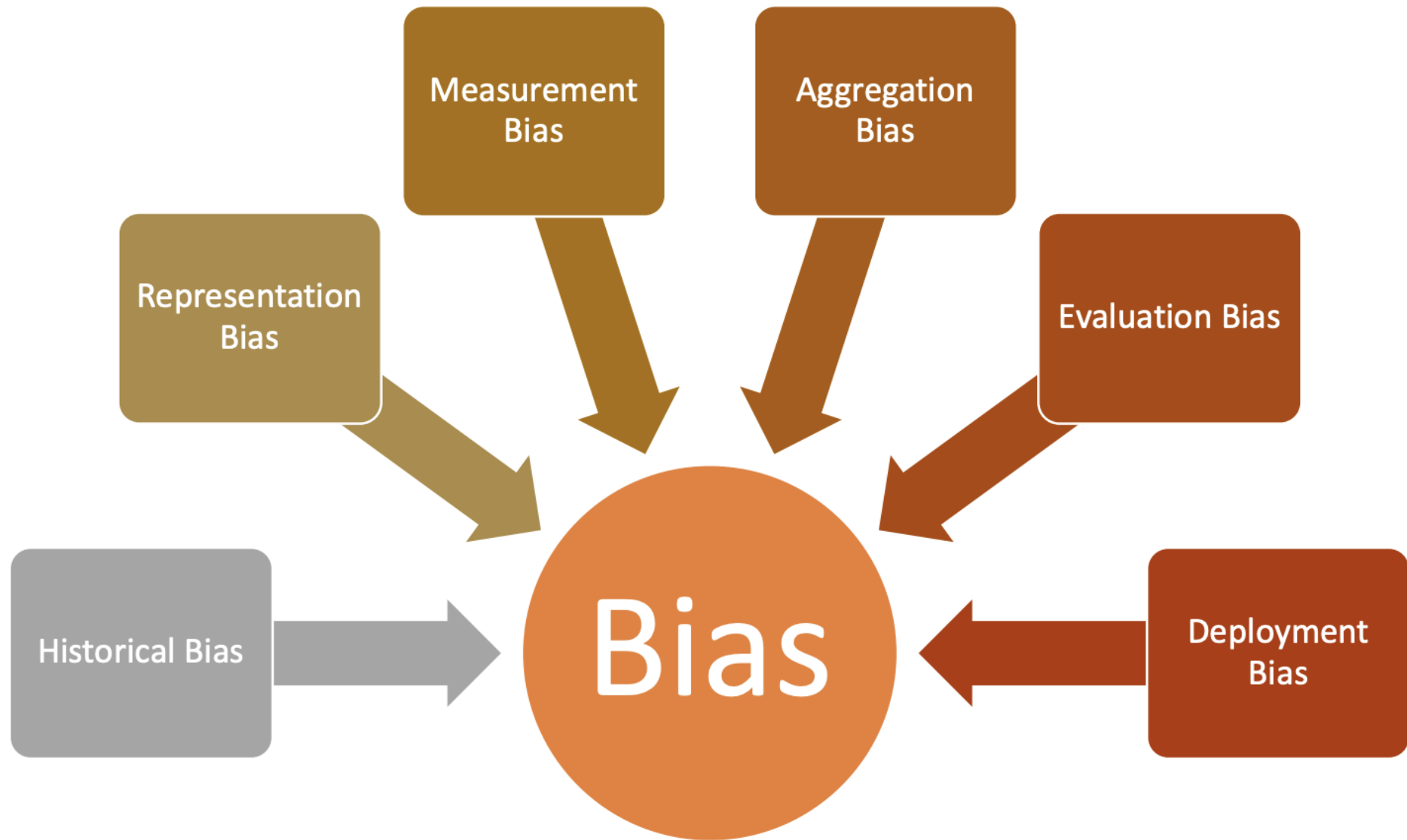
An abstract background on the left side of the slide. It features a dark, textured area with a grid of small, colorful dots in shades of red, orange, yellow, and green. Overlaid on this is a large, semi-transparent, stylized shape that resembles a hand or a cluster of fingers, composed of many small, overlapping circles in the same color palette.

A key focus of this course...

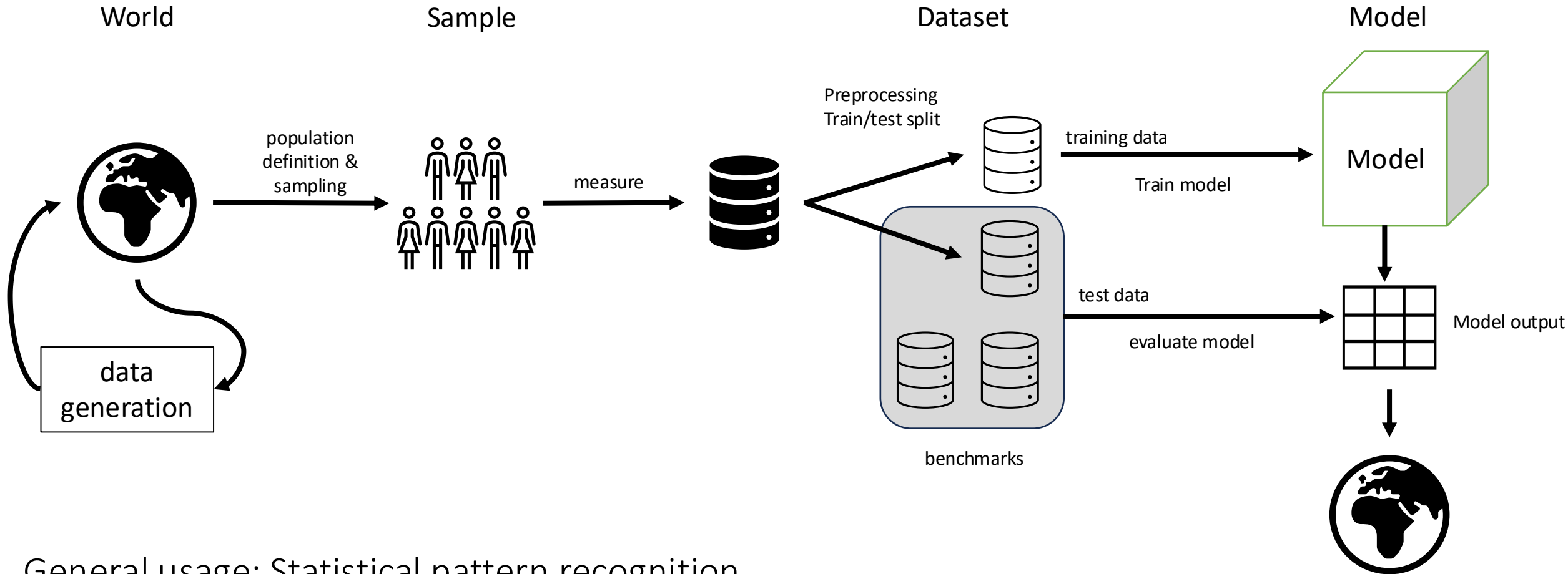
“computer systems that systematically and unfairly discriminate against certain individuals or groups of individuals in favor of others.”

Bias in Computer Systems - Friedman & Nissenbaum, 1996

Types of Bias

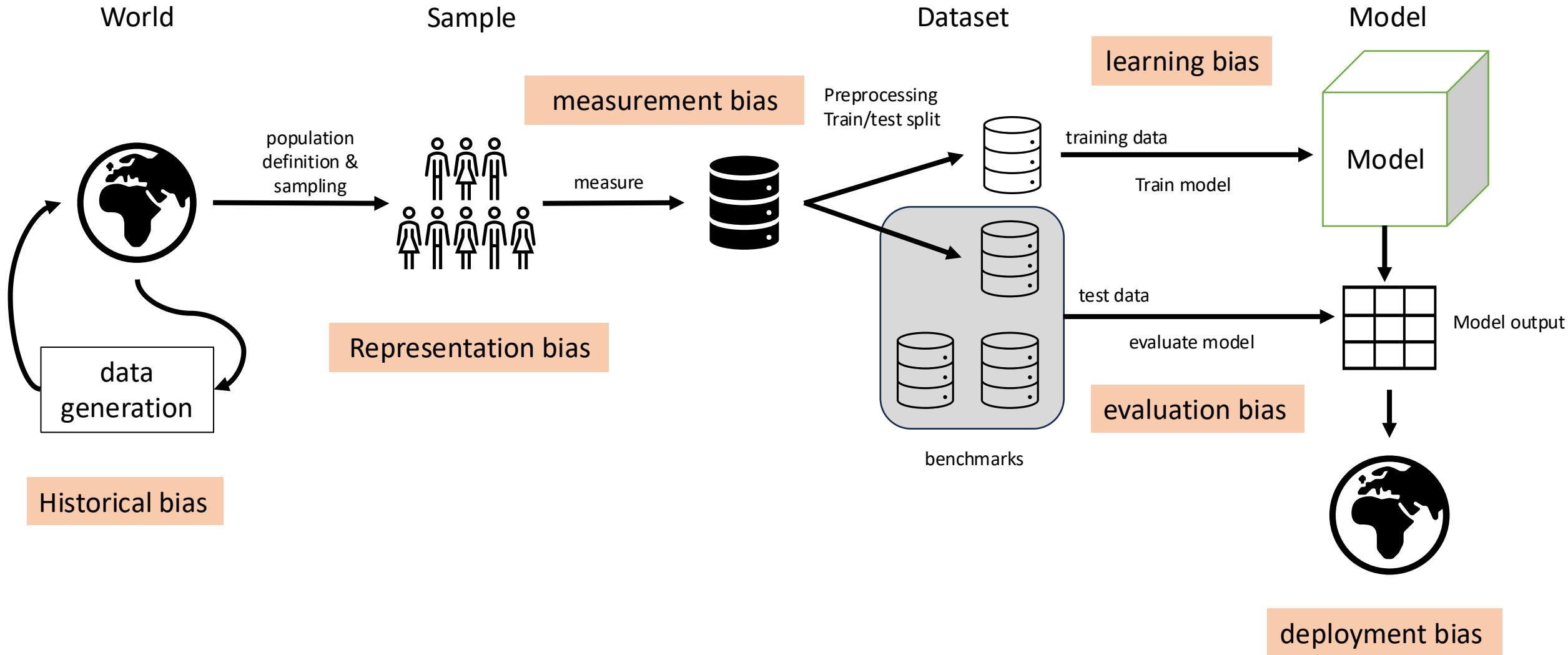


AI systems



General usage: Statistical pattern recognition,
trained over vast data sets

Bias can be introduced at multiple points in the pipeline



Historical bias

- Historical and or social prejudices and beliefs are reflected in data. When there is a misalignment between world as it is and the values or objectives to be encoded and propagated in a model.
- It is a normative concern with the state of the world and exists even given perfect sampling and feature selection.
- Recognizing historical bias requires a retrospective understanding of the application and data generation process over time.

Interesting read: “Word embeddings quantify 100 years of gender and ethnic stereotypes”:
<https://www.pnas.org/content/115/16/E3635>

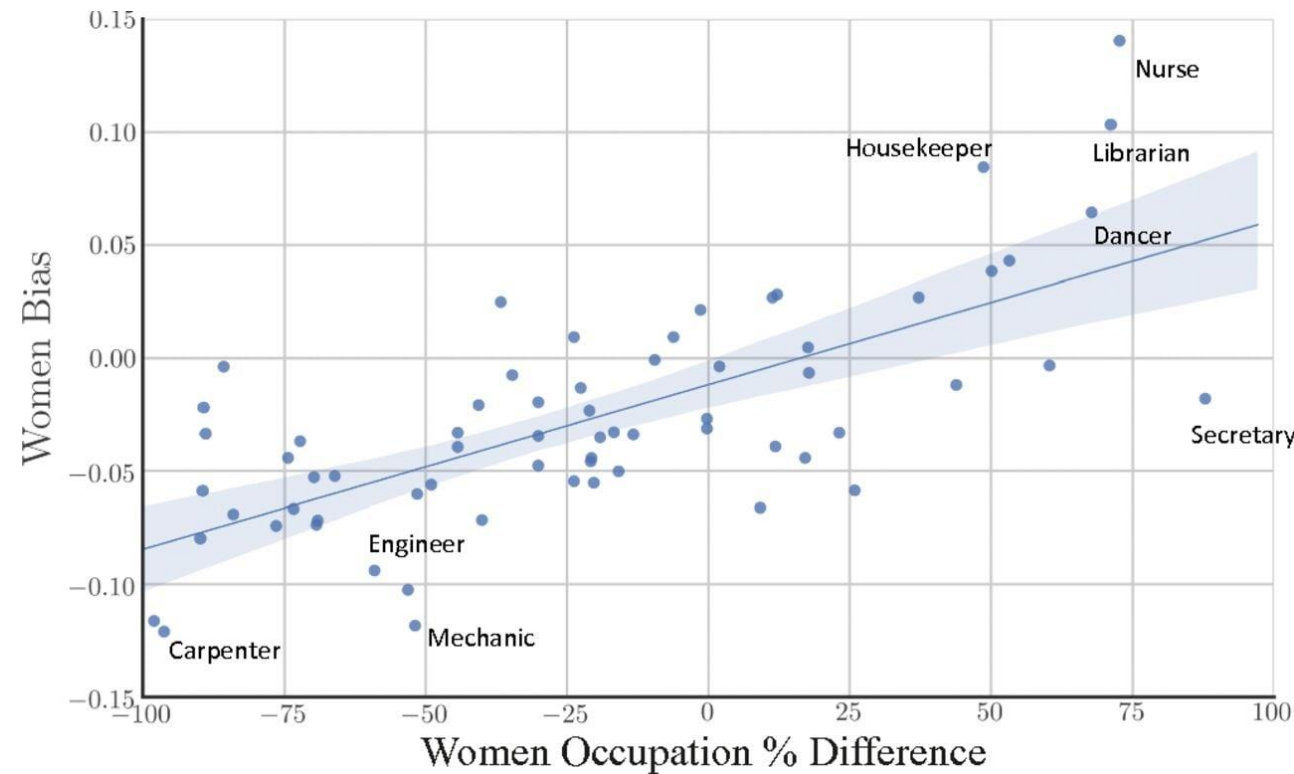


Word embeddings quantify 100 years of gender and ethnic stereotypes

Examine gender bias in word embeddings trained on Google News data (occupations) and compare Women's occupation relative percentage vs. embedding bias in Google News vectors. →

More positive indicates more associated with women on both axes.

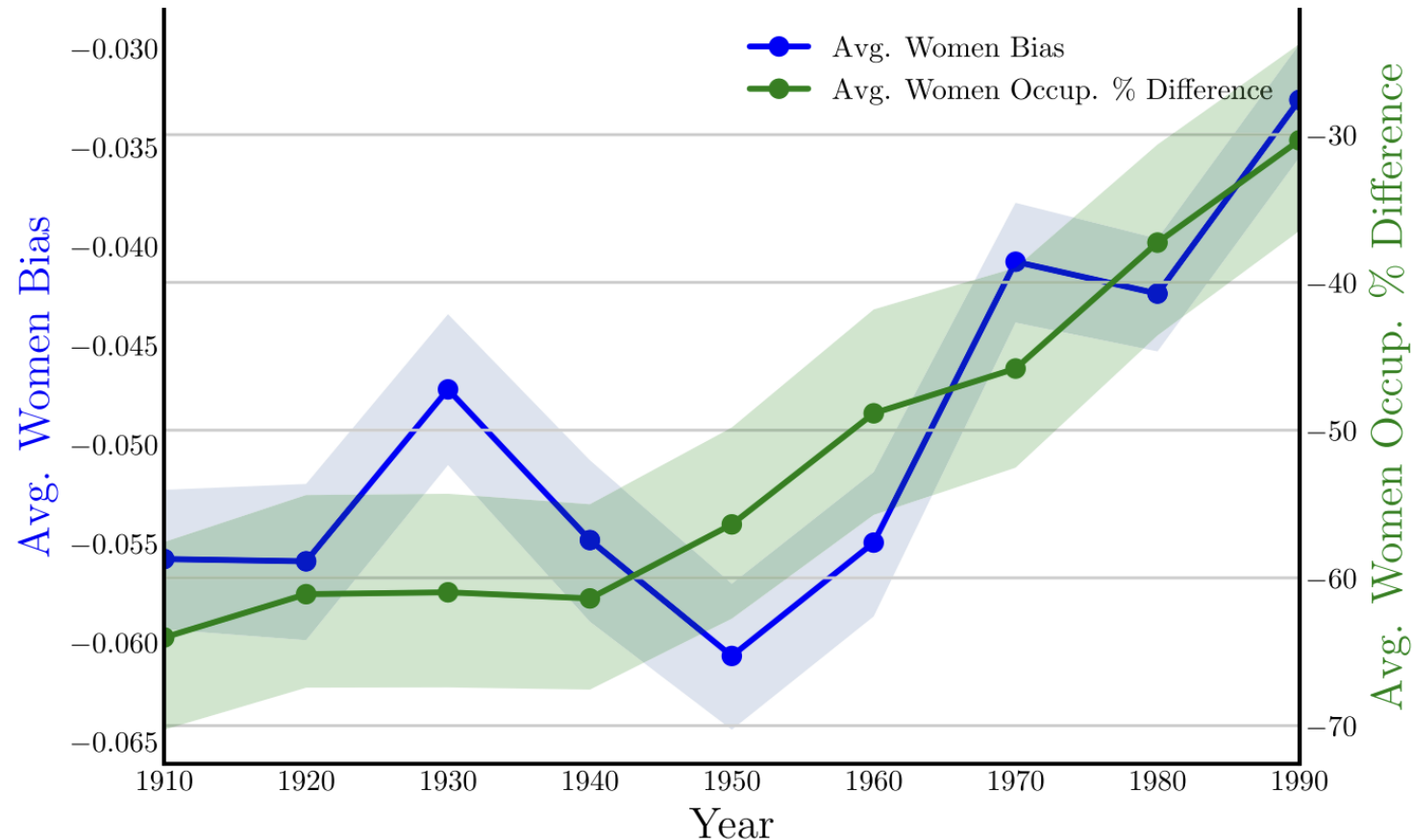
The association in the embedding effectively captures the percentage of women in an occupation.



Source: Garg et al, 2018

<https://www.pnas.org/content/115/16/E3635>

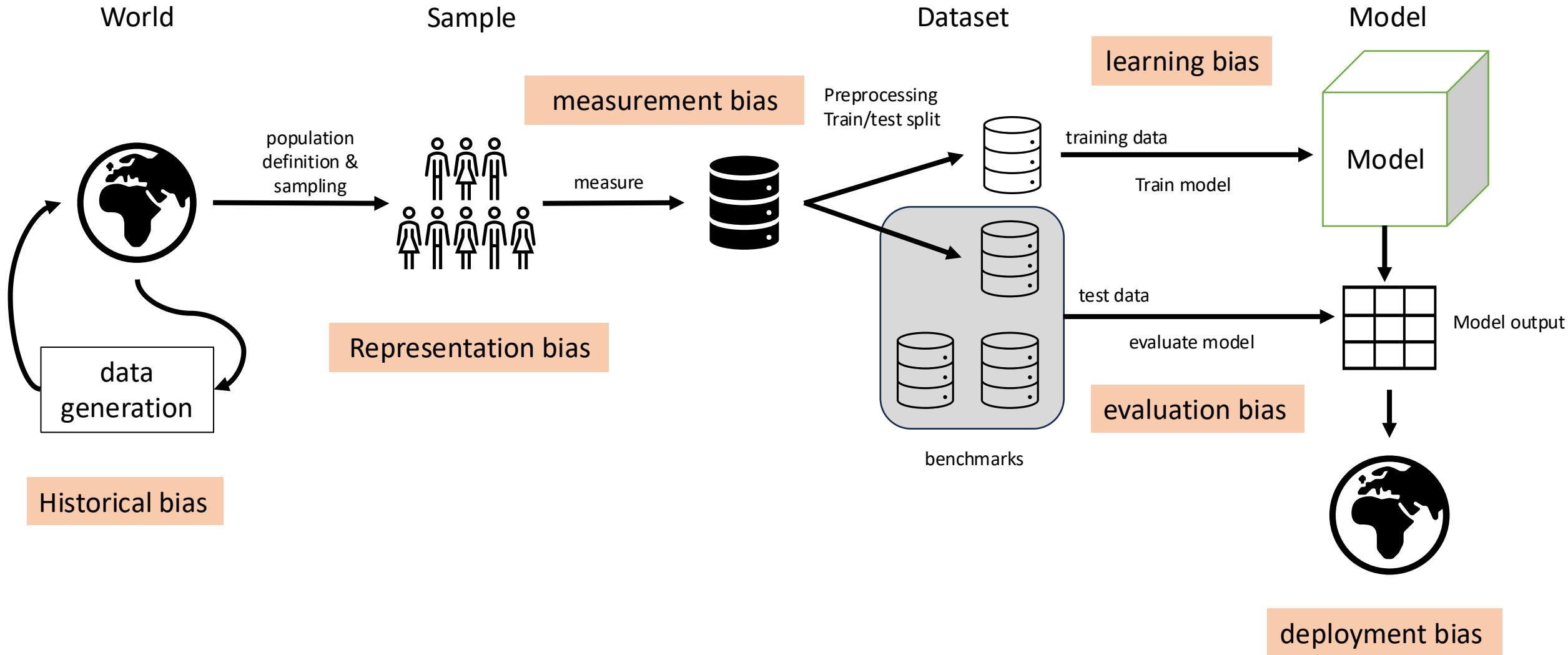
Word embeddings quantify 100 years of gender and ethnic stereotypes



Historical bias

- Using data that doesn't reflect current realities
 - E.g. occupation data on what roles were historically over/under-represented by different demographic groups but that have leveled out such as medicine
- Failing to account for systemic discrimination that has shaped modern distribution
 - E.g. Lack of gender and ethnic diversity in leaderships positions, e.g. CTOs.
 - E.g. financial exclusion
- Failing to align with modern *values*
 - E.g. natural language datasets that terms no longer in use, have changed meaning or that have since been acknowledged as offensive or harmful

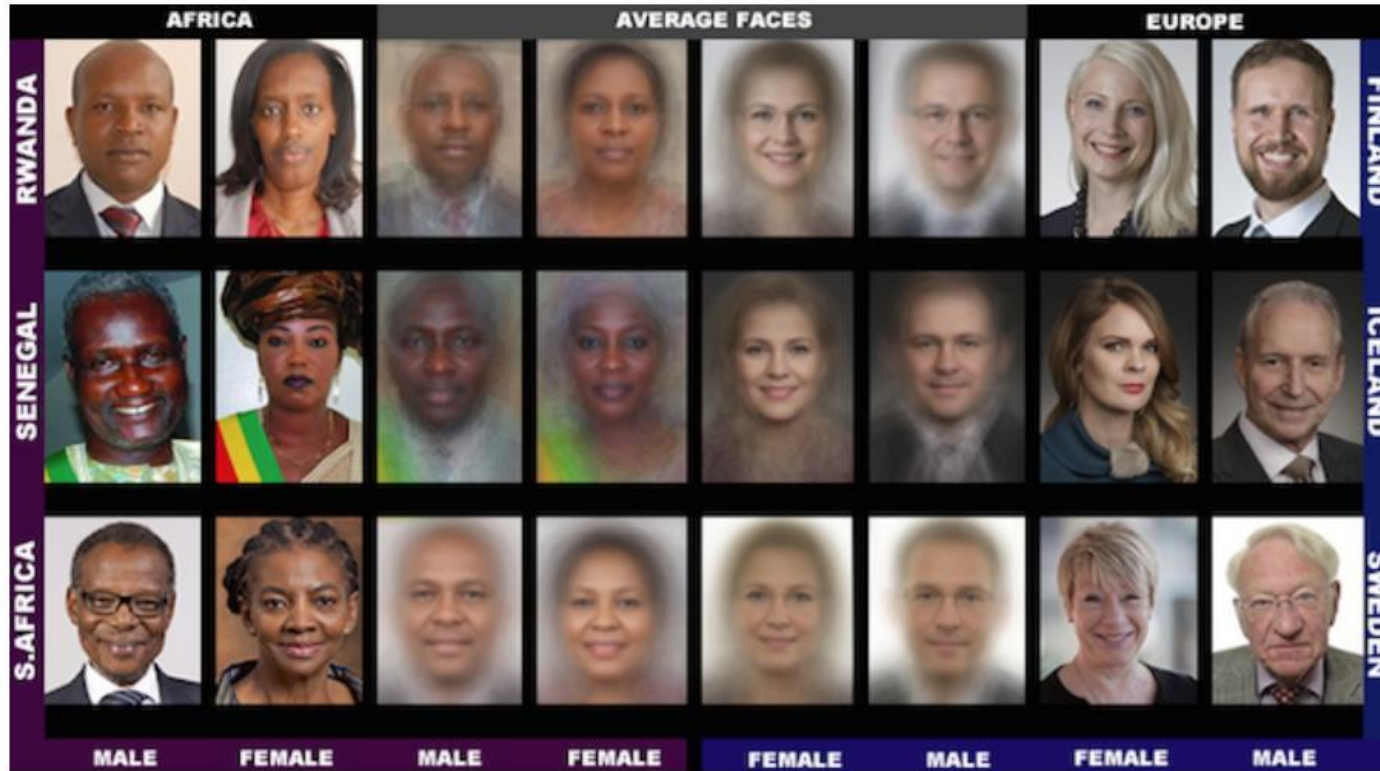
Bias can be introduced at multiple points in the pipeline



Representation bias

- Arises while defining and sampling a population.
- Technically speaking...
 - It occurs when the development population under-represents, and subsequently fails to generalize well, for some part of the use population.

Gender Shades (Buolamwini & Gebru, 2018)

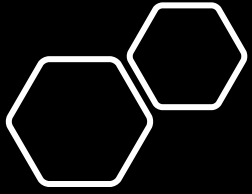


- Benchmark datasets used in facial recognition had 79.6% and 86.2% light-skinned subjects.

Pilot Parliaments Benchmark

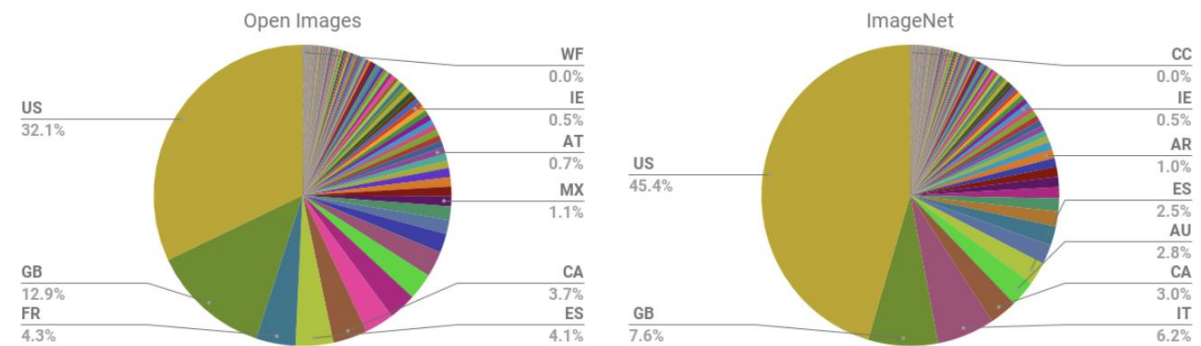
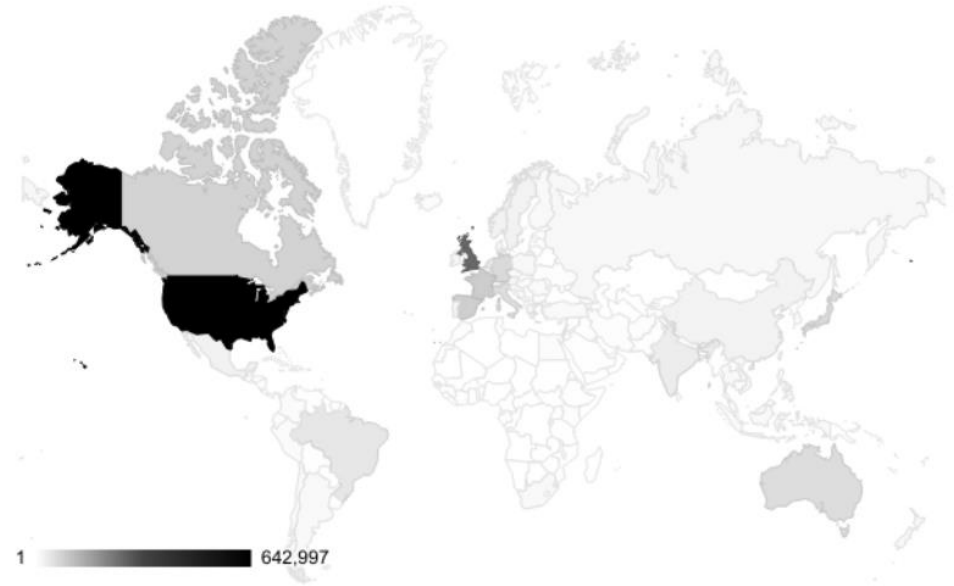
Image credit: Joy Buolamwini

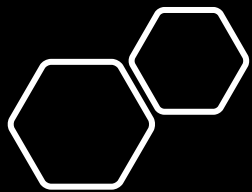
<https://youtu.be/TWWsW1w-BVo>



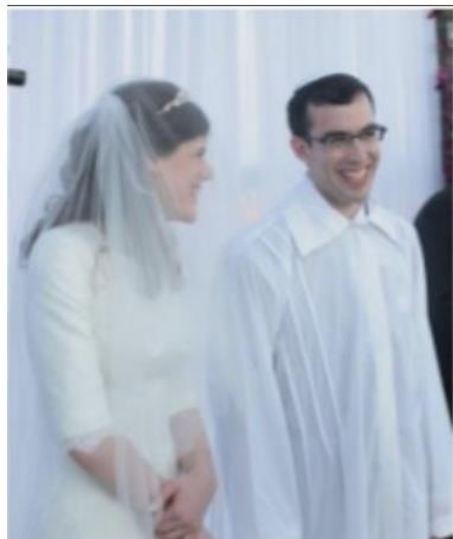
Geographical representation in image datasets

- Open Image and ImageNet
- One third of the data was collected in US
- 60% of the data was from the six most represented countries.

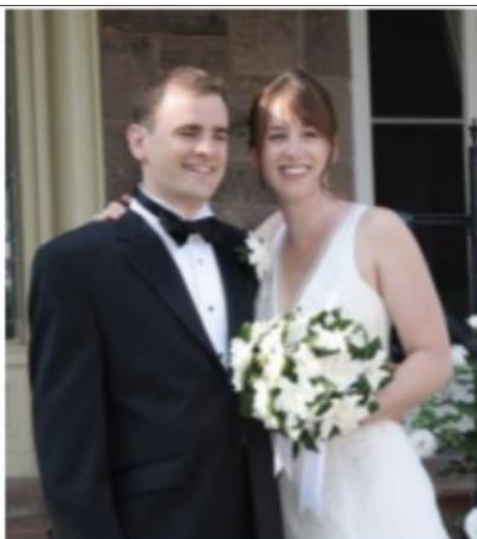




Annotation bias: image labels



*ceremony,
wedding, bride,
man, groom,
woman, dress*



*ceremony,
bride, wedding,
man, groom,
woman, dress*



person, people

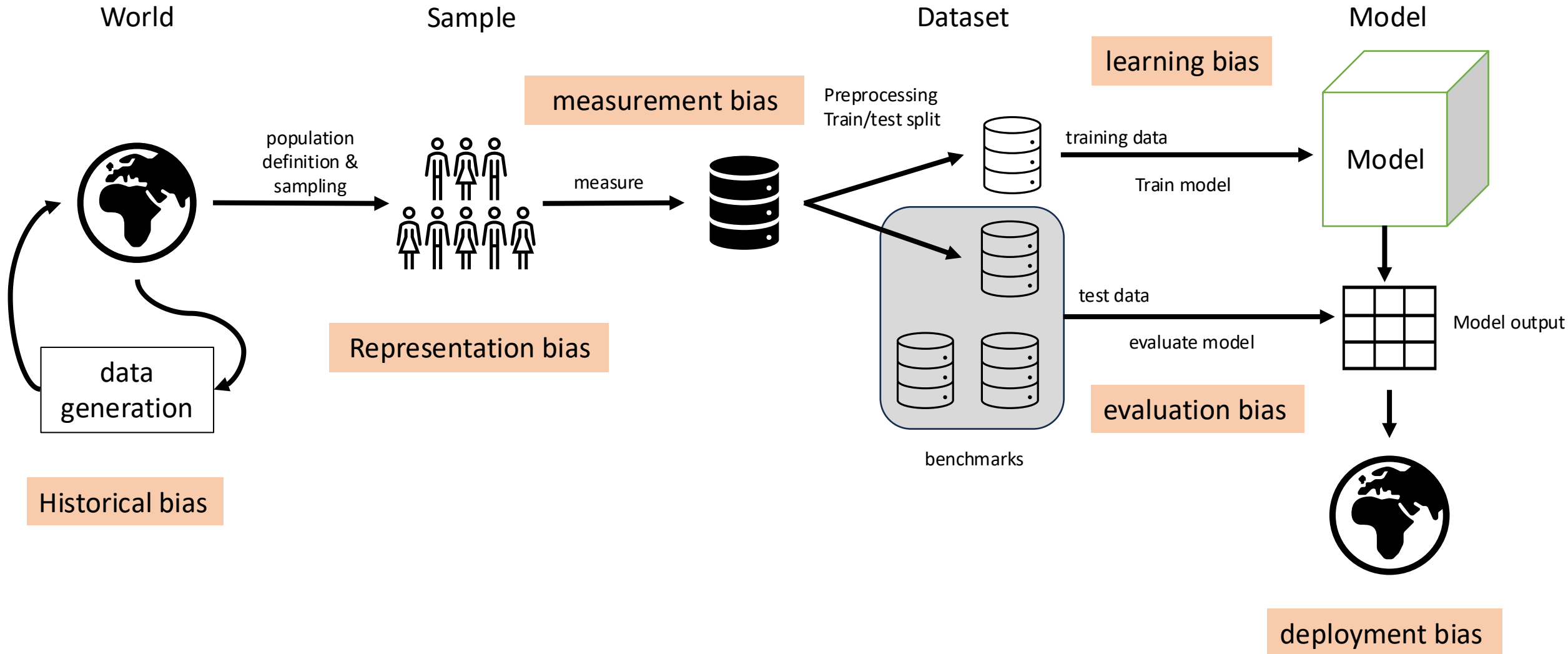
Annotation bias: toxicity & hatespeech detection

- Subjective assessment of what is offensive
- African American dialect more likely to be labeled as toxic / hate speech (e.g. Sap et al)
- Can be improved via training, added contextual information



\$ # ! ?

Bias can be introduced at multiple points in the pipeline



Data: “What gets counted counts”

“Data must be classified in some way to be put to use. In fact, by the time that information becomes data, it’s already been classified in some way. Data, after all, is information made *tractable*.”

(D’Ignazio & Klein, 2021)

Measurement bias

- Arises when choosing and measuring features and labels to use
- In data that is generated by human decision makers, one often sees human decisions used as proxies,
 - For example, “arrested” is used as a proxy for “crime,” or “pain medication prescribed by doctor” is used as a proxy for “patient’s pain.”
 - These are often proxies for the desired quantities.
- The chosen set of features and labels may leave out important factors or introduce group or input-dependent noise that leads to differential performance.

Example...

If we think of an algorithm that classifies candidates as 'hireable' or not, or loan applicants creditworthy – it is necessary to decide what data should contribute to that assessment.

- If you were designing an algorithm to assess 'hireability' or 'creditworthiness' what data would you use?
- What categories are in our data – are they observable or created?
- Is there potential for bias?

Aggregation bias

- Especially when distinct populations are inappropriately combined.
- In many applications, the population of interest is heterogeneous, and a single model is unlikely to suit all subgroups.
- Identifying aggregation bias usually requires some understanding of meaningful groups and reason to think they have different conditional distributions with respect to the prediction label.
- Medical applications, for example, often risk aggregation bias because patients of different genders with similar underlying conditions present and progress in different ways.

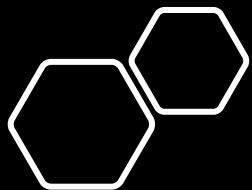
Deployment bias

- a mismatch between the problem a model is intended to solve and the way in which it is actually used
- May occur when built and evaluated as though it were entirely self-contained
- In reality: embedded within complex sociotechnical system, inc. institutional structures and human decision-makers

→ Sound familiar? Similar to emergent bias – as described by Friedman & Nissenbaum?!

Types of bias: Summary

- **Historical bias** occurs when the state of the world in which the data was generated is flawed.
- **Representation bias** occurs when building datasets for training a model, if those datasets poorly represent the people that the model will serve.
- **Measurement bias** occurs when the accuracy of the data varies across groups. This can happen when working with proxy variables (variables that take the place of a variable that cannot be directly measured), if the quality of the proxy varies in different groups.
- **Aggregation bias** occurs when groups are inappropriately combined, resulting in a model that does not perform well for any group or only performs well for the majority group. (most commonly arises in medical applications, e.g. where ethnicity or gender may impact on likelihood of certain conditions).
- **Evaluation bias** occurs when evaluating a model, if the benchmark data (used to compare the model to other models that perform similar tasks) does not represent the population that the model will serve.
- **Deployment bias** occurs when the problem the model is intended to solve is different from the way it is actually used. If the end users don't use the model in the way it is intended, there is no guarantee that the model will perform well.



Countering bias

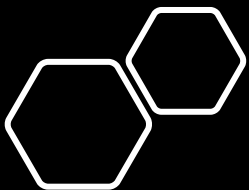
Mindful selection and sampling of data.

Checking error rates and measuring fairness and conducting audits of existing data

Annotation guidelines and contextual information

Efforts to include groups that may be mis-or under-represented in historical datasets

Engagement with groups that will be affected by a model or technology to understand impacts and potential harms



Countering bias



Mindful selection and sampling of data

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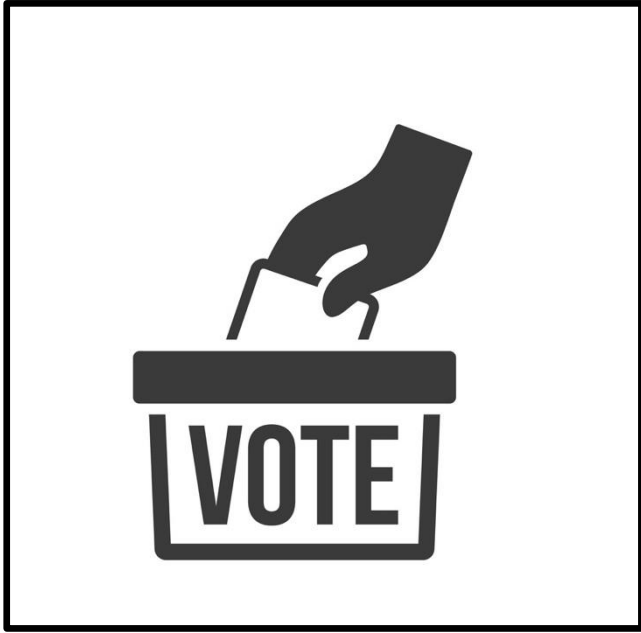
Engagement with groups that will be affected by a model or technology to understand impacts and potential harms



What does it mean for a machine learning model to be 'fair'?

General definition: *impartial and just treatment or behaviour without favouritism or discrimination.*





What do we mean by fair?

Procedural fairness: is the process to determine and outcome fair?
e.g. coin toss, voting



Outcome fairness: is the distributions of outcomes perceived as fair?
e.g. sharing out dinner servings based on hunger/need, Salary increase based on performance

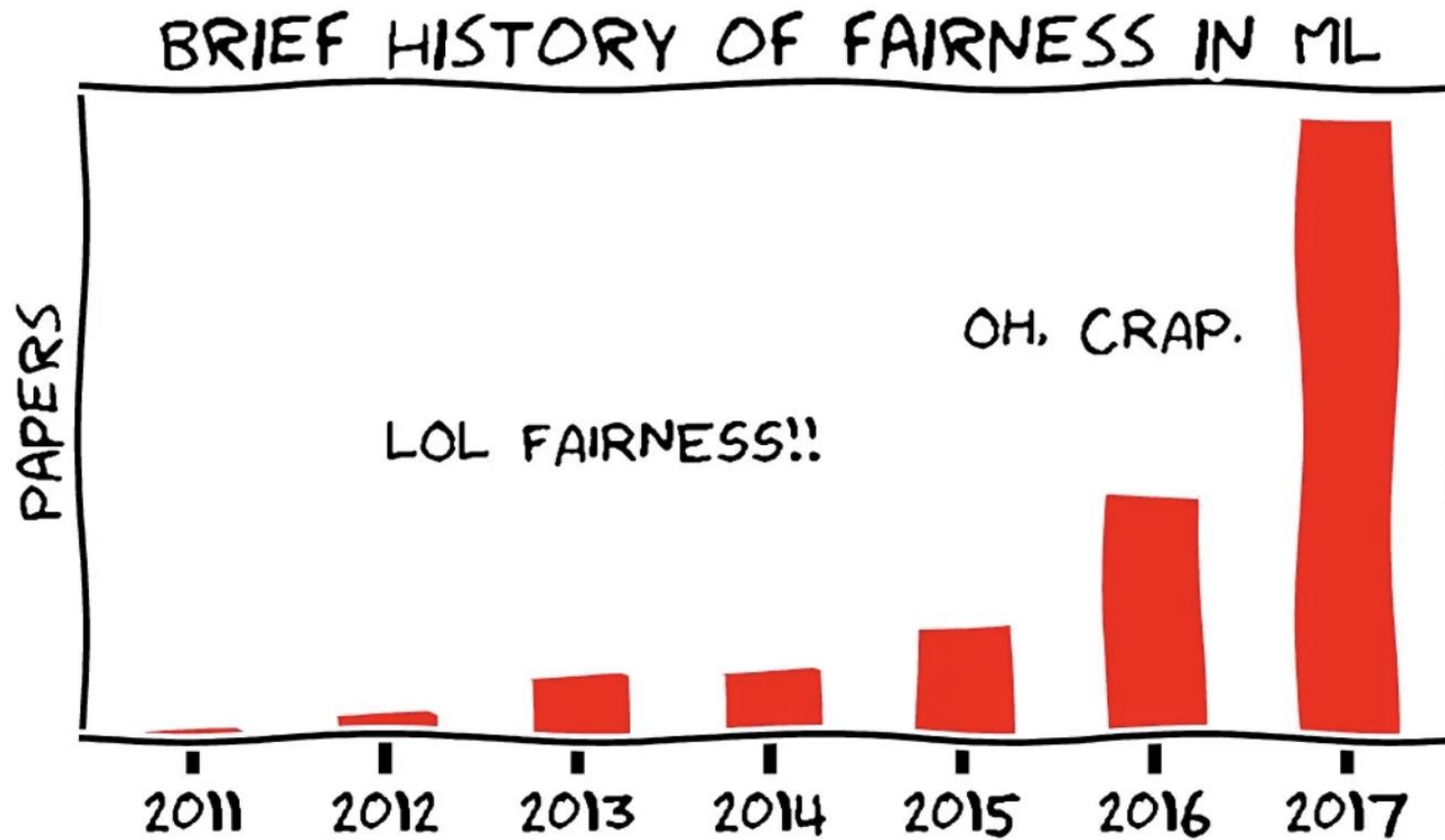


Equality



Equity

Fairness in machine learning



How to quantify fair model performance?

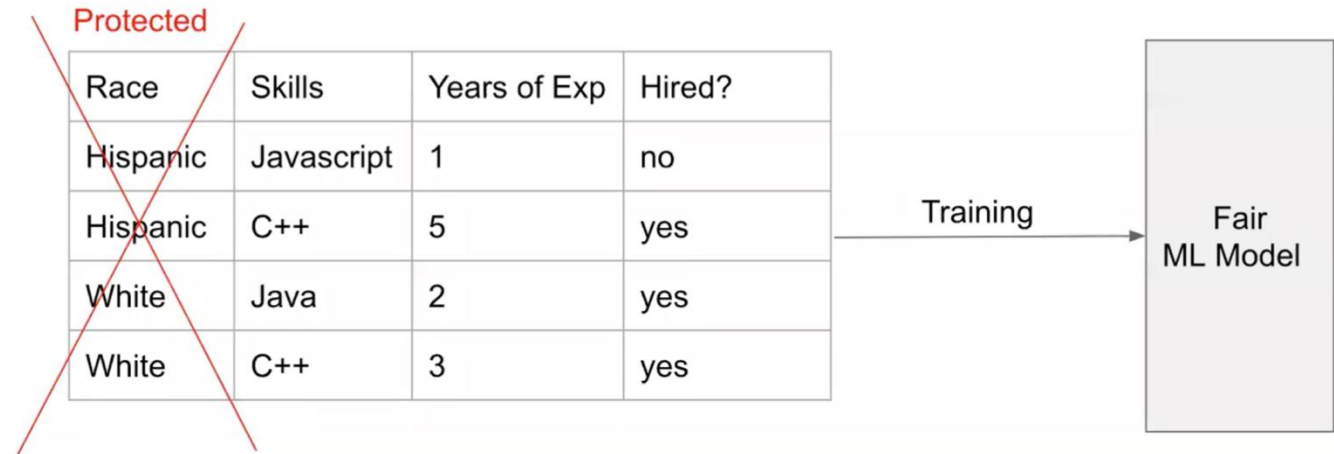
Group fairness: criteria to compare quantities at a group level, typically identified by sensitive attributes (e.g. gender, ethnicity, age, etc.)

We might say...

- The performance of the model for all groups should be the same
- The performance of the model in the lowest performing group should be as high as possible
- The variation in performance across diverse groups should be as low as possible

Fairness through unawareness?

- It might be tempting to create a model that ignores sensitive characteristics.
- E.g, simply drop all data that explicitly identifies which group each candidate belongs to
- The model should be 'unaware' that the subgroups even exist,
- Should make discrimination impossible... right?

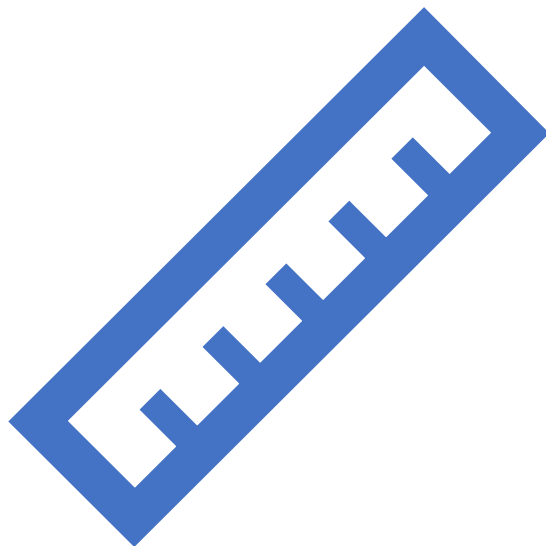


Fairness through unawareness?

- Removing a sensitive feature doesn't work if the model can reconstruct it from other features in the data
- It is often possible to infer group membership through other data (profiling, redundant encodings etc)

- Sensitive features may still be used
 - Inferred from indirect evidence





How to measure fairness?



Defining fairness

Various different ways to define and measure fairness, e.g.

- Demographic parity: equal positive prediction rates across groups
- Equalized odds / equal opportunity: equal true positive and/or false positive rates across groups
- Calibration: predicted scores mean the same thing regardless of group

→ Once you pick a fairness definition, you can test for it through error analysis.

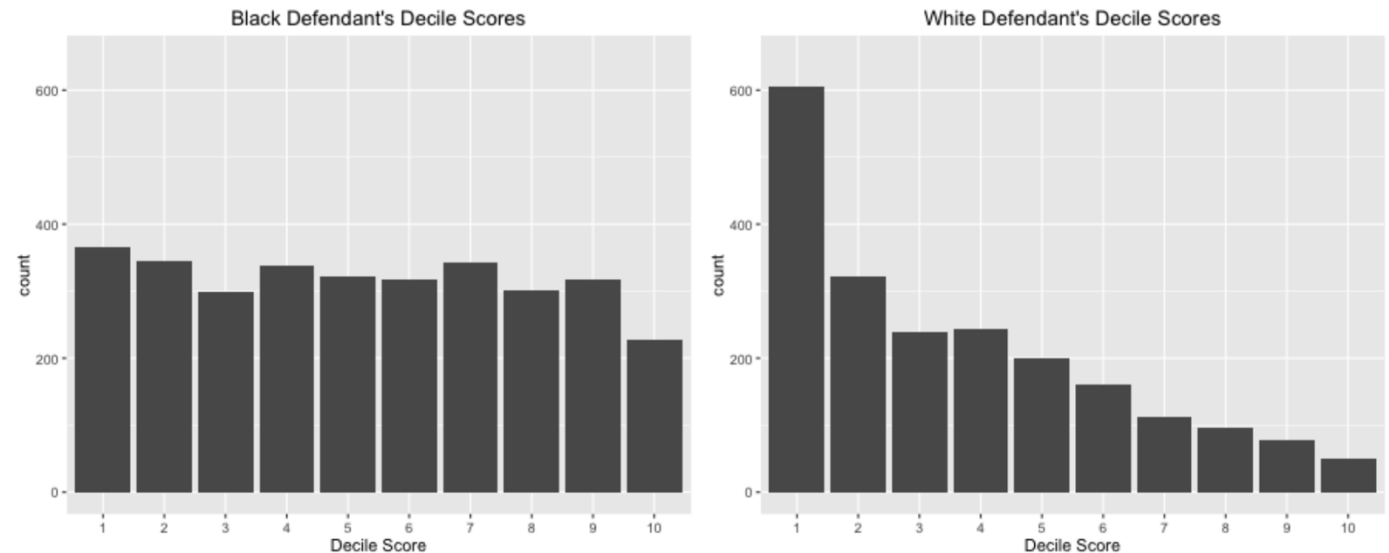
Case Study: COMPAS

1. Person commits a crime, is arrested
2. COMPAS software predicts the likelihood that the person will commit another crime in the future (*recidivism*)
3. Recidivism scores impact criminal sentences: if a person is likely to commit another crime, shouldn't they get a longer sentence?

→ Real system that has been used in New York, Wisconsin, California, Florida, etc

Case Study: COMPAS

2016 ProPublica article analyzed COMPAS scores for >7000 people arrested in Broward county, Florida



Question: How many of these people ended up committing new crimes within 2 years?

Error Analysis

- Let's imagine we have a predictive model classify an event
- The model predicts either YES or NO
- Matrix of outcomes:

		True result	
		YES	NO
Predicted result	YES	True positive (tp)	False positive (fp)
	NO	False negative (fn)	True negative (tn)

Error Analysis: example COMPAS

- Is a defendant at high/low risk of reoffending?

- Error Rate: $FP + FN / TP + FP + FN + TN$
 - How often is the prediction wrong?
- False Positive Rate: $FP / (FP + TN)$
 - How often were non-offenders predicted to reoffend?
- False Negative Rate: $FN / (FN + TP)$
 - How often were offenders predicted not to reoffend?

	Outcome: Recidivism	Outcome: No recidivism
Prediction: High risk	True Positive (TP)	False Positive (FP)
Prediction: Low risk	False Negative (FN)	True Negative (TN)

Error Metrics: Different Stakeholders

	Outcome: Recidivism	Outcome: No recidivism
Prediction: High risk	True Positive (TP)	False Positive (FP)
Prediction: Low risk	False Negative (FN)	True Negative (TN)

$$\text{Error Rate} = \frac{FP + FN}{TN + FP + FN + TP}$$

How often is the prediction wrong?

Defendants care about this

$$\text{False Positive Rate} = \frac{FP}{FP + TN}$$

How often were non-offenders predicted to reoffend?

Judges care about this

$$\text{False Negative Rate} = \frac{FN}{FN + TP}$$

How often were offenders predicted not to reoffend?

Error Metrics: Results

	Reoffend	Did not reoffend
High risk	True Positive (TP)	False Positive (FP)
Low risk	False Negative (FN)	True Negative (TN)

$$\text{Error Rate} = \frac{FP + FN}{TN + FP + FN + TP} \approx 34.6\%$$

$$\text{False Positive Rate} = \frac{FP}{FP + TN} \approx 32.4\%$$

$$\text{False Negative Rate} = \frac{FN}{FN + TP} \approx 37.4\%$$

Error Metrics: Results by group

Black defendants	Reoffend	Did not reoffend
High risk	1369 (TP)	805 (FP)
Low risk	532 (FN)	990 (TN)

Error Rate $\approx$ 36.2%

White defendants	Reoffend	Did not reoffend
High risk	505 (TP)	349 (FP)
Low risk	461 (FN)	1139 (TN)

Error Rate $\approx$ 33.0%

Similar error rates between groups

Error Metrics: Results by group

Black defendants	Reoffend	Did not reoffend
High risk	1369 (TP)	805 (FP)
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Error Rate $\approx 36.2\%$

False Positive Rate $\approx 44.9\%$

White defendants	Reoffend	Did not reoffend
High risk	505 (TP)	349 (FP)
Low risk	461 (FN)	1139 (TN)

Error Rate $\approx 33.0\%$

False Positive Rate $\approx 23.5\%$

Black defendants have 1.9x higher False Positive Rate!

Error Metrics: Results by group

Black defendants	Reoffend	Did not reoffend
High risk	1369 (TP)	805 (FP)
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Error Rate $\approx 36.2\%$

False Positive Rate $\approx 44.9\%$

False Negative Rate $\approx 28.0\%$

White defendants	Reoffend	Did not reoffend
High risk	505 (TP)	349 (FP)
Low risk	461 (FN)	1139 (TN)

Error Rate $\approx 33.0\%$

False Positive Rate $\approx 23.5\%$

False Negative Rate $\approx 47.7\%$

White defendants have 1.7x higher False Negative Rate

Error Metrics: Results by group

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Metrics

Source – Wikipedia:
https://en.wikipedia.org/wiki/Sensitivity_and_specificity

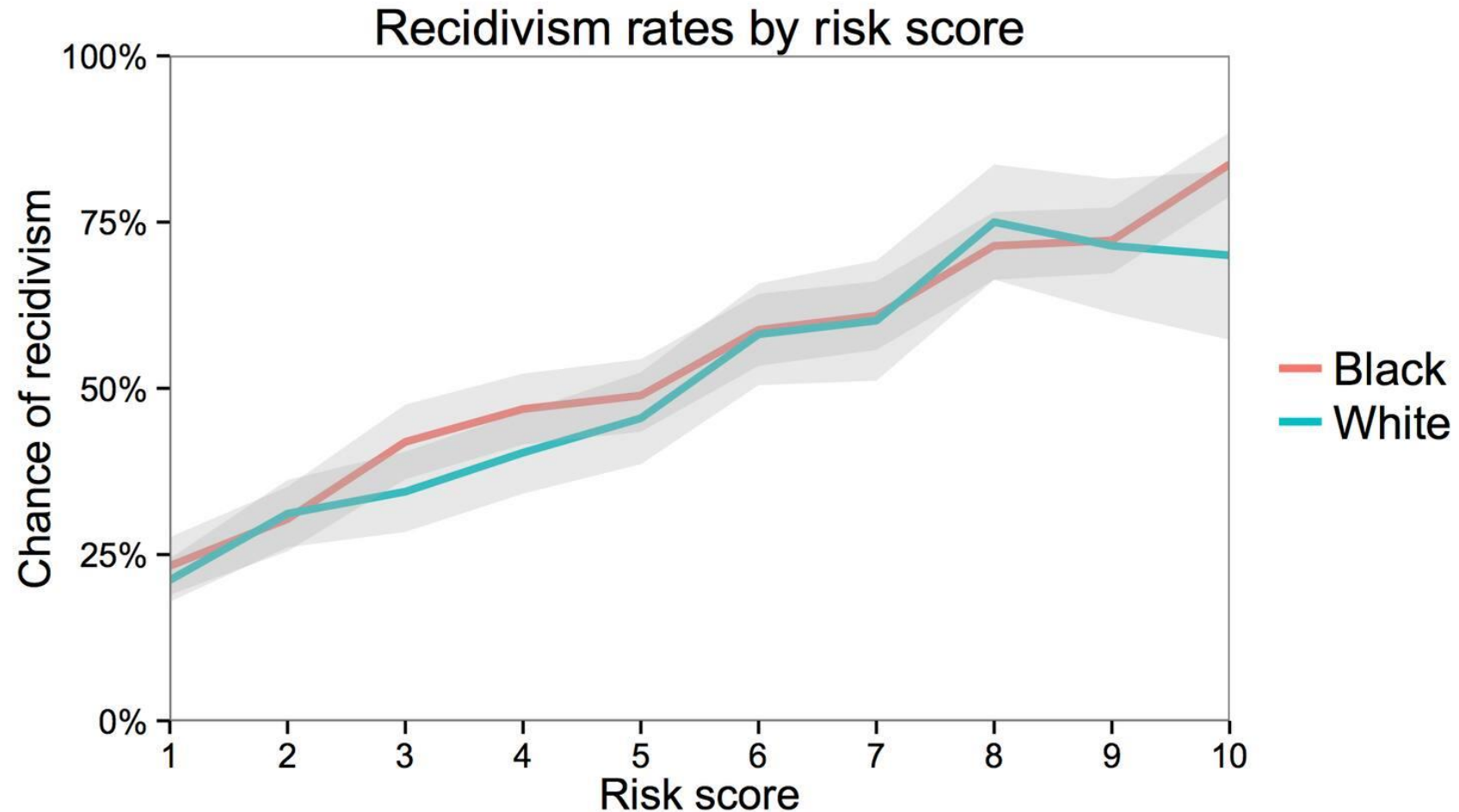
		Predicted condition		Sources: [16][17][18][19][20][21][22][23] view · talk · edit	
Actual condition	Total population = P + N	Predicted positive	Predicted negative	Informedness, bookmaker informedness (BM) = TPR + TNR - 1	Prevalence threshold (PT) $= \frac{\sqrt{\text{TPR} \times \text{FPR}} - \text{FPR}}{\text{TPR} - \text{FPR}}$
	Real Positive (P) [a]	True positive (TP), hit ^[b]	False negative (FN), miss, underestimation	True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate, power $= \frac{\text{TP}}{\text{P}} = 1 - \text{FNR}$	False negative rate (FNR), miss rate type II error ^[c] $= \frac{\text{FN}}{\text{P}} = 1 - \text{TPR}$
	Real Negative (N) [d]	False positive (FP), false alarm, overestimation	True negative (TN), correct rejection ^[e]	False positive rate (FPR), probability of false alarm, fall-out type I error ^[f] $= \frac{\text{FP}}{\text{N}} = 1 - \text{TNR}$	True negative rate (TNR), specificity (SPC), selectivity $= \frac{\text{TN}}{\text{N}} = 1 - \text{FPR}$
	Prevalence $= \frac{\text{P}}{\text{P} + \text{N}}$	Positive predictive value (PPV), precision $= \frac{\text{TP}}{\text{TP} + \text{FP}} = 1 - \text{FDR}$	False omission rate (FOR) $= \frac{\text{FN}}{\text{TN} + \text{FN}}$ = 1 - NPV	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$
		False discovery rate (FDR) $= \frac{\text{FP}}{\text{TP} + \text{FP}} = 1 - \text{PPV}$	Negative predictive value (NPV) $= \frac{\text{TN}}{\text{TN} + \text{FN}}$ = 1 - FOR	Markedness (MK), deltaP (Δp) = PPV + NPV - 1	Diagnostic odds ratio (DOR) $= \frac{\text{LR}+}{\text{LR}-}$
		F ₁ score $= \frac{2 \text{ PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}}$ $= \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}}$	Fowlkes– Mallows index (FM) $= \sqrt{\text{PPV} \times \text{TPR}}$	phi or Matthews correlation coefficient (MCC) $= \frac{\sqrt{\text{TPR} \times \text{TNR} \times \text{PPV} \times \text{NPV}}}{\sqrt{\text{FNR} \times \text{FPR} \times \text{FOR} \times \text{FDR}}}$	Threat score (TS), critical success index (CSI), Jaccard index $= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$

Fair? It depends what measure you use

Northpoint's claim -
Calibration: A score of 7 means the same thing regardless of race.

ProPublica's claim —
equal error rates

Both claims are correct. The conflict is mathematically inevitable.



COMPAS & Fairness measures



Demographic parity: equal positive prediction rates across groups



Equalized odds / equal opportunity: equal true positive and/or false positive rates across groups

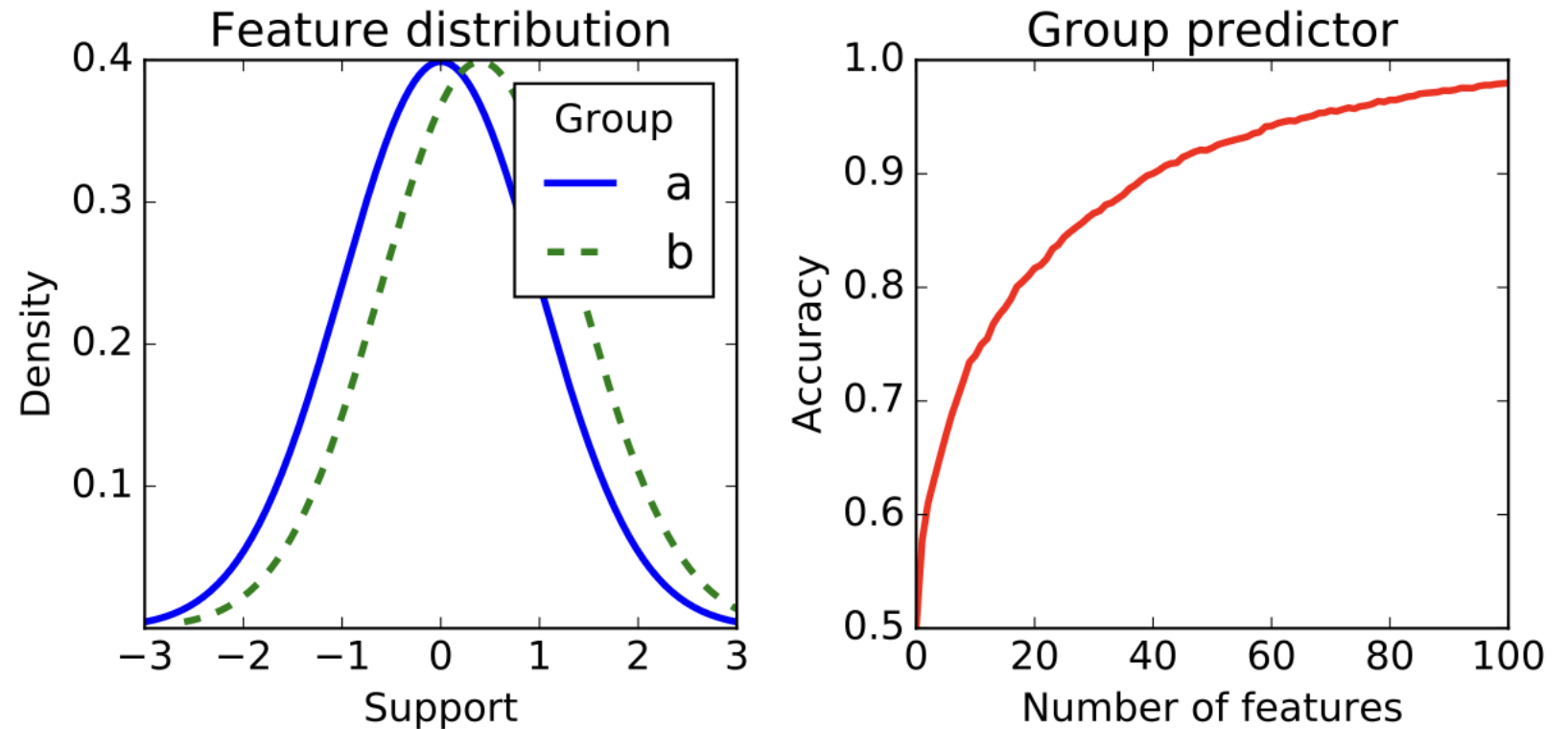


Calibration: predicted scores mean the same thing regardless of group

The real question is therefore not technical — it's ethical: which errors matter more, and who gets to decide?

No fairness through unawareness

COMPAS scores defendants 1–10 on reoffending risk using 100+ factors. Race is excluded, but scores still differ by race because correlated factors vary by race.



Fairness is a complicated topic

- **At least 21** mathematical definitions of fairness
- **At least 71** metrics to detect bias in machine learning
- It is proved that we cannot satisfy all (or even part) of them simultaneously.
- Which definition of fairness to use depends on the use-case

No perfect fairness measure

- it is mathematically impossible to satisfy all definitions of group fairness simultaneously
- Necessary to communicate with end-users and stakeholders to determine how best to train models and audit their performance.

Fairness

- **At least 21** mathematical definitions of fairness
- **At least 71** metrics to detect bias in machine learning
- Cannot satisfy all (or even part) of them simultaneously.

Source: Barocas, S., Hardt, M., & Narayanan, A. (2017). Fairness in machine learning. *Nips tutorial*, 1, 2017.

List of demographic fairness criteria			
Name	Closest relative	Note	Reference
Statistical parity	Independence	Equivalent	Dwork et al. (2011)
Group fairness	Independence	Equivalent	
Demographic parity	Independence	Equivalent	
Conditional statistical parity	Independence	Relaxation	Corbett-Davies et al. (2017)
Darlington criterion (4)	Independence	Equivalent	Darlington (1971)
Equal opportunity	Separation	Relaxation	Hardt, Price, Srebro (2016)
Equalized odds	Separation	Equivalent	Hardt, Price, Srebro (2016)
Conditional procedure accuracy	Separation	Equivalent	Berk et al. (2017)
Avoiding disparate mistreatment	Separation	Equivalent	Zafar et al. (2017)
Balance for the negative class	Separation	Relaxation	Kleinberg, Mullainathan, Raghavan (2016)
Balance for the positive class	Separation	Relaxation	Kleinberg, Mullainathan, Raghavan (2016)
Predictive equality	Separation	Relaxation	Chouldechova (2016)
Equalized correlations	Separation	Relaxation	Woodworth (2017)
Darlington criterion (3)	Separation	Relaxation	Darlington (1971)
Cleary model	Sufficiency	Equivalent	Cleary (1966)
Conditional use accuracy	Sufficiency	Equivalent	Berk et al. (2017)
Predictive parity	Sufficiency	Relaxation	Chouldechova (2016)
Calibration within groups	Sufficiency	Equivalent	Chouldechova (2016)
Darlington criterion (1), (2)	Sufficiency	Relaxation	Darlington (1971)

In summary

When producing such metrics...

1. Choose metrics that quantify what you are interested in
2. Present them carefully, mindful of how an audience may interpret them.
3. Understand that no single metric can tell the whole story.

In summary

- Inspect the rationale and design decisions
 - E.g. target variables - how defined / computed
- Inspect the data
 - who is represented, is demographic data available?
 - What labels are used, how were annotations generated?
- Conduct error testing
 - Use multiple differential performance measures
 - Utilize existing fairness measures and bias metrics (numerous toolkits available)
- Test against specialist benchmark datasets

Limitations & Implications

- Multiple measures of fairness
- Different measures benefit different stakeholders
- Error metrics may focus on outcome but not procedural fairness
- Group fairness relies on the availability of demographic data on sensitive attributes / protected characteristic
 - Not always available...
- Group fairness measures can fail to account for intersectionality

Summary

- We want to make not just the models fair, but the outcomes as well.
 - It's not good enough to choose a model that you believe should be fair, you must compare this against its outputs
- Important to assess and communicate to stakeholders about the types of bias resulting from the data, models, etc., so that:
 - Models can be tuned to meet the need of system users AND impacted groups
 - System users can make informed decisions about how to act on model outputs
 - Transparency / Accountability

Reading on bias & harms

- To consolidate this week's learning:
 - Read: Suresh, H., & Gutttag, J. (2021). A framework for understanding sources of harm throughout the machine learning life cycle. In Equity and access in algorithms, mechanisms, and optimization (pp. 1-9).
 - Watch: Tutorial: 21 fairness definitions and their politics: https://www.youtube.com/watch?v=jlXluYdnyyk&feature=emb_logo

Additional resources:

- Fair Machine learning text-book (Chapters 1-4)
 - Online textbook for the course: Barocas, Solon, Moritz Hardt, and Arvind Narayanan. Fairness and Machine Learning , 2018, Available online: <https://fairmlbook.org/>

Feedback

- Demographic parity
- Equalised odds
- Predictive rate parity